Lemma. Let G be a group and A be a set.
For an Action of G on A, $|G:G_a| = |C_a|$

Proof. Since for any $g_1, g_2 \in G_a$, $g_1 g_2^{-1} a = g_1 a = a$, so $g_1 g_2^{-1} \in G_a$.

Thus G_a is a subgroup. And, define a map

$$C_a \to \{gG_a \mid g \in G\} ; g \cdot a \mapsto gG_a$$

Then, $g_1 G_a = g_2 G_a$ implies $g_1^{-1} g_2 \in G_a$, or $g_1^{-1} g_2 a = a$, or $g_1 a = g_2 a$.
So injective, and the surjective is trivial. Thus, Proved.

Thm. Let G be a group and H be a subgroup of G.
An action $G \times \{gH \mid g \in G\} \to \{gH \mid g \in G\} ; (g_1, g_2 H) \mapsto g_1 g_2 H$
satisfies:
1) It acts transitively. (It is clear, since $\{gH \mid g \in G\}$ is a Partition of G).
2). $G_H = H$
3) $\ker \pi_H = \bigcap_{x \in G} xHx^{-1}$ is the largest normal subgroup of G contained in H.

Proof. $\ker \pi_H = \{g \in G \mid gxH = xH \text{ for all } x \in G\}$
$= \{g \in G \mid x^{-1} g x H = H \text{ for all } x \in G\} = \bigcap_{x \in G} xHx^{-1}$.
Let $g \in G$ and $h \in \ker \pi_H$. Then, for any $x \in G$,
$x^{-1}(ghg^{-1})x = (g^{-1}x)^{-1} \cdot h \cdot (g^{-1} \cdot x) \in \ker \pi_H$, so $\ker \pi_H \trianglelefteq G$.

And, given $g \in \ker \pi_H$ and $h \in H$, $h^{-1} g h H = H \Rightarrow g \in H$, so $\ker \pi_H \leq H$.
Finally, if $N \leq H$ and $N \trianglelefteq G$, then given $x \in G$

$$N = xNx^{-1} \leq xHx^{-1}, \text{ so } N \subseteq \bigcap_{x \in G} xHx^{-1}.$$

Corollary. If G is a finite group of order n and p is the smallest prime dividing n, then any subgroup of index p is normal.
Proof. Let $H \leq G$ such that $|G:H| = p$. And, let $K = \ker \pi_H$, π_H is an action above.
Since $K \leq H$, let $|H:K| = k$. Then, $|G:K| = |G:H| \cdot |H:K| = p \cdot k$.
Since H has p left cosets, $G/K \cong \pi_H[G] \leq S_p$, so $pk = |G/K|$ divides $p!$,
so $k = 1$, $H = K$.

「Automorphism of Group.

「Prop. Let H be a normal subgroup of the group G.
Then, G acts by conjugation on H as automorphisms of H.
More specifically, for each $g \in G$,

$$H \to H : h \mapsto ghg^{-1}$$

is an automorphism of H. Moreover, the permutation represent,

$$\pi_H : G \to S_H : g \mapsto \sigma_g \qquad \text{where } \sigma_g(h) = ghg^{-1}$$

is a Homomorphism of G into $Aut(H) \leq S_H$, with $\ker \pi_H = C_G(H)$.
In Summary, $G/\ker \pi_H = G/C_G(H) \cong Im\, \pi_H \leq Aut(H)$, and $G/Z(G) \cong Inn(G)$.

Proof. Since H is normal, the Action is well-defined. And,

$$\ker \pi_H = \{g \in G \mid \sigma_g = id\} = \{g \in G \mid ghg^{-1} = h \text{ for all } h \in H\} = C_G(H).$$

「Corollary. For any subgroup H of G,
$N_G(H)/C_G(H)$ is isomorphic to a subgroup of $Aut(H)$.

「Prop. $Aut(\mathbb{Z}_n) \cong \mathbb{Z}_n^\times$.
Proof. Denote $\mathbb{Z}_n = \langle x \rangle$, where $x \in \mathbb{Z}_n$. Given automorphism $\varphi \in Aut(\mathbb{Z}_n)$,
the φ is uniquely determined by $\varphi(x) = x^a$, the integer $1 \leq a \leq n$,
and $|x| = |\varphi(x)| = |x^a|$, so $\gcd(a,n) = 1$. Now,

$$Aut(\mathbb{Z}_n) \to \mathbb{Z}_n^\times ; \varphi_a \mapsto a \bmod n$$

is the isomorphism.

$$\mathbb{Z}_8^\times = \{1, 3, 5, 7\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Note: If P is a group of a prime order p, then

$$Aut(P) \cong \mathbb{Z}_p^\times \cong \mathbb{Z}_{p-1}.$$

⌈Lemma. Let $P \in Syl_p(G)$. If Q is p-subgroup of G, then $Q \cap N_G(P) = Q \cap P$.

proof. Since $P \subseteq N_G(P)$, $Q \cap P \subseteq Q \cap N_G(P)$ is trivial.

Since $Q \cap N_G(P) \le N_G(P)$, denote $H = Q \cap N_G(P)$,

$$PH = \bigcup_{h \in H} Ph = \bigcup_{h \in H} hP = HP$$

Thus $PH \le G$ and $|PH| = \dfrac{|P| \cdot |H|}{|P \cap H|}$. By the Lagrange's Thm, $H \le Q$ means order of H is power of p. Therefore, $P \le PH$ and $|PH|$ is power of p, thus $PH = P$, or $H \le P$. Now, $H = Q \cap N_G(P) \subseteq Q$, $H \le Q \cap P$. ∎

⌈Lemma. If G is finite abelian group and p is a prime number dividing $|G|$, then G has an element of order of p.

Proof. Using induction for $|G|$. If $|G| = 1$, trivial.

Now, suppose that the statement hold for all groups whose order is smaller than $|G|$.

Since $|G| > 1$, take $x \in G$ with $x \neq 1$. That is, $|x| > 1$.

If $|G| = p$, then $|x| = p$, by the Lagrange's Theorem $(|x| \mid p)$.

Now, suppose $|G| \neq p$. $(|G| > p)$.

If p divides $|x|$, $|x| = pn$ for some $n \in \mathbb{N}$, so $|x^n| = p$. So clear.

If p does not divides $|x|$, let $N = \langle x \rangle$. (Clearly, $p \nmid |N| = |x|$).

Since G is abelian, N is Normal, and $x \neq 1$, so it is non-trivial.

By the Lagrange Theorem, $|G| = |G/N| \cdot |N|$. But, p does not divide $|N|$, p divides $|G/N|$. Now, using the hypothesis of the induction G/N has an element of order p, denote this $\bar{y} = yN$. Since $\bar{y} \neq \bar{1}$, $y \notin N$.

But, $\bar{y}^p = \bar{1}$, so $y^p \in N$. This implies $\langle y^p \rangle \subsetneq \langle y \rangle$. Since $|y^p|$ divides $|y| = p$ the only possibilities are $|y^p| = 1$, this y is we desired.

「The Sylow Theorem. Let G be a group of order $p^n \cdot m$, p is prime not dividing m.

「First. $Syl_p(G) \neq \emptyset$.

「Second. If $P \in Syl_p(G)$ and $Q \leq G$ is a p-subgroup, then $Q \leq gPg^{-1}$ for some $g \in G$.

「Third. $n_p \equiv 1 \mod p$, $n_p = |G : N_G(P)|$ for any $P \in Syl_p(G)$, and $n_p | m$.

Proof. To show the first, we will use the induction for order of G.
If $|G| = 1$, then trivial. Now, suppose that $Syl_p(H) \neq \emptyset$, for all group whose order is less than G.
In the Case of p divides the order $Z(G)$, then by the Cauchy's Theorem for abelian group, there is an order p element y in $Z(G)$. Put $N = \langle y \rangle$. Then, N is Normal of G.
Now, G/N is group of order of $p^{n-1} \cdot m$, and by the hypothesis of induction, G/N has a Subgroup of order of p^{n-1}, denote this $\overline{P} = P/N < G/N$.
By the 4th Isomorphism Theorem, there is $P < G$ s.t $|P/N| = p^{n-1}$, so $|P| = p^n$.
In the Case of p does not divide the order of $Z(G)$,
let $g_1, \cdots, g_r$ be the distinct non-central representatives of Conjugacy classes of G.
Then, by the class equation,

$$p^n \cdot m = |G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)]$$

Then, for some $1 \leq j \leq r$, p does not divides $[G : C_G(g_i)]$, because $p \nmid |Z(G)|$.
For this j, let $H = C_G(g_j)$. Since $g_j \notin Z(G)$, $|H| < |G|$.
Now, since $[G : H]$ has no factor of p, so $|H| = p^\alpha \cdot k$, $p \nmid k$, $k < m$
By the hypothesis of the induction H has a subgroup of order of p^α,
so this is a subgroup we desired.

「Corollary. Let $P \in Syl_p(G)$.
$n_p = 1 \iff P \trianglelefteq G \iff P$ char G

⌈Group of order of pq ($p<q$ are prime, $p \nmid q-1$).

If $Z(G) \neq 1$, then $G/Z(G) \cong \mathbb{Z}_p$ or $\mathbb{Z}_q$ by the Lagrange Theorem, then G must be abelian, since $G/Z(G)$ is cyclic.

Now, suppose that $Z(G)=1$.

By the Lagrange Thm, every non-identity element of G has order p or q.

If there is no order q element, then the class equation gives

$$pq = 1 + \sum |G : C_G(g_i)| = 1 + kq,$$

because $C_G(g_i) = \langle g_i \rangle \cong \mathbb{Z}_p$. (Since $g_i \notin Z(G)$, $C_G(g_i) \not> \langle g_i \rangle$).

This is contradiction, since $q \nmid 1$. So, G has a order q element, denote this to x.

Let $H = \langle x \rangle$. Since $|G:H| = p$ is the smallest prime dividing $|G|$, so H is normal.

Now, $G/H = N_G(H)/C_G(H)$ is a subgroup of order p, and isomorphic to a subgroup of $Aut(H)$.

Since $Aut(H) \cong \mathbb{Z}_q^\times$ has order of $q-1$, this implies $p \mid q-1$, contradiction.

「Group of order of pq, $p<q$. (p, q are prime)

Let $P \in Syl_p(G)$ and $Q \in Syl_p(G)$.

Since n_q divides p, $n_q = 1$ or p, but $n_q \equiv 1 \bmod q$, thus $n_q = 1$. $\therefore Q \trianglelefteq G$.

1. Case of $p \nmid q-1$.

By the Sylow Thm, $n_p = 1$ or q, but if $n_p = q$, then $q \equiv 1 \bmod p$, $p \mid q-1$, Contradiction.

Therefore, $n_p = 1$, $P \trianglelefteq G$. Now, G must be a cyclic, because;

Since $|P|$ and $|Q|$ are primes, both are Cyclic. Let $P = \langle x \rangle$ and $Q = \langle y \rangle$.

Since $P \trianglelefteq G$, $N_G(P) = G$. Therefore,

$$G/C_G(P) = N_G(P)/C_G(P) \cong S \text{ for some } S \leq Aut(\mathbb{Z}_p)$$

But, $|Aut(\mathbb{Z}_p)| = p-1$, thus S is trivial. (because $|S|$ divides $|G| = pq$).

Now, $G = C_G(P)$, and $x \in P$, x and y are Commute. This implies $|xy| = pq$. $G = \langle pq \rangle$.

2. Case of $p \mid q-1$

Let Q be a Sylow q-subgroup of S_q, the symmetric group q letters.

Since q is prime, $|N_{S_q}(Q)| = q \cdot (q-1)$, because

1). Since $|S_q| = q!$, $|Q| = q$.

2). S_q has $(q-1)!$ q-Cycles.

3). Since order of gQg^{-1} is q, these are all Cyclic.

4). $gQg^{-1} \cap Q = \{1\}$ for all $g \in G$, so

$$n_q = \#\{gQg^{-1} \mid g \in G\} = (q-1)!/(q-1) = (q-2)!$$

therefore, $n_q = G_Q = [S_q : N_G(Q)] = (q-2)!$, or $N_{S_q}(Q) = q(q-1)$. 」

Now, Since $p \mid q-1$, p divides order of $N_{S_q}(Q)$, so it has a subgroup $\overset{P}{\text{?}}$ of order p.

And, since $P \leq N_{S_q}(Q)$, PQ is subgroup of order pq.

And, $C_{S_q}(Q) = Q$, PQ is non-abelian group.

$$|G : N_{S_q}(Q)| = G_Q$$

「Group of order 48 is <u>Not</u> simple.

$48 = 2^4 \cdot 3$. Therefore,

$$n_3 = 1 \text{ or } 4 \text{ or } 16 \quad \text{and} \quad n_2 = 1 \text{ or } 3.$$

If $n_2 = 1$, then $P \in Syl_2(G)$ is the Normal, so there is nothing to proven

Suppose $n_2 \neq 1$, or $n_2 = 3$. That is, there are three Sylow 2-subgroup,

$$P_1, P_2, P_3 \subseteq G. \quad (\# P_i = 16).$$

For each $1 \leq i < j \leq 3$,

$$|P_i P_j| = \frac{|P_i| \cdot |P_j|}{|P_i \cap P_j|} = \frac{2^8}{|P_i \cap P_j|} \leq 48 = |G| \qquad \text{if } k,\ P_i = P_j$$

So, $\frac{2^4}{3} \leq |P_i \cap P_j|$, $|P_i \cap P_j|$ cannot be $1, 2, 4, 16$ therefore

$$|P_i \cap P_j| = 8.$$

Take $Q = P_1 \cap P_2$, as subgroup of order 8. Since $Q \subseteq P_1, P_2$ and index 2,

$Q \trianglelefteq P_1$ and $Q \trianglelefteq P_2$. That is, for any $g \in P_1 \cup P_2$,

$$gQg^{-1} = Q.$$

Therefore, $P_1 \cup P_2 \subseteq N_G(Q)$. Since $|P_1 \cap P_2| = 8$,

$$|P_1 \cup P_2| = |P_1| + |P_2| - |P_1 \cap P_2| = 24$$

Now, if $|N_G(Q)| = 24$, then it has index 2 to G,

$$N_G(Q) \trianglelefteq G$$

And, if $|N_G(Q)| = 48$, then $N_G(Q) = G$, so $Q \trianglelefteq G$.

「Group of order 30 is Not simple.

$30 = 2 \cdot 3 \cdot 5$, let $P \in Syl_5(G)$ and $Q \in Syl_3(G)$.

If either P or Q is normal, then $PQ \leq G$, and it is group of order 15, because $|PQ| = \frac{|P| \cdot |Q|}{|P \cap Q|} = 15$. (All elements of P has order 5 except identity, and " Q " 3 ").

So, PQ is index of 2 to G, so PQ is normal subgroup.

Now, assume neither P nor Q is normal. By the Sylow Theorem, $n_5 = 6$ and $n_3 = 10$. By the Lagrange's Theorem, for any $P_1, P_2 \in Syl_5(G)$, $P_1 \cap P_2 = \{1\}$. So, there are $6 \cdot 4 = 24$ distinct order 5 elements. Similarly, there are 20 distinct order 3 elements, $24 + 20 = 44 \leq 30$, Contradiction. Therefore, either P or Q must be normal, so PQ is index 15 subgroup.

In sum. group of order 30 must have a subgroup of order of 15.

「Group of order 105 is Not simple.

$105 = 3 \cdot 5 \cdot 7$, so

$$n_3 = 1 \text{ or } 7 \qquad n_5 = 1 \text{ or } 21 \qquad n_7 = 1 \text{ or } 15$$

If it is simple, then $n_3 = 7$, $n_5 = 21$, $n_7 = 15$. Therefore there are

$$\underbrace{2}_{\text{order } 3} \cdot 7 + \underbrace{4}_{\text{order } 5} \cdot 21 + \underbrace{6}_{\text{order } 7} \cdot 15 = 14 + 84 + 90 \geq 130$$

So, Contradiction by Counting

⌈Group of order 12 is Not Simple.

Every group of order 12 has a normal Sylow 3-Subgroup or $G \cong A_4$.

$12 = 2^2 \cdot 3$. If $n_3 = 1$, then G is not simple, it has a normal sylow 3-subgroup.

Suppose that $n_3 \neq 1$. Then, there is only possibility is $n_3 = 4$.

Let $P \in Syl_3(G)$. By the Sylow Theorem, $|G : N_G(P)| = n_3 = 4$,

so $P = N_G(P)$, by $P \subseteq N_G(P)$ and $|N_G(P)| = 12/4 = 3$, given by the Lagrange Thm.

Now, Consider an action of G on $Syl_3(G)$ by Conjugation:

$$G \times Syl_3(G) \to Syl_3(G) : (g, P) \mapsto gPg^{-1}$$

Since $Syl_3(G) = \{P_1, P_2, P_3, P_4\}$, we can write the permutation representation of this action:

$$\varphi : G \to S_4 : g \mapsto \sigma_g \quad \text{where} \quad \sigma_g : S_4 \to S_4 : P_i \mapsto gP_ig^{-1}$$

Now, the elements of $\ker \varphi$ normalizes all sylow 3-subgroups. That is,

$$\ker \varphi \leq N_G(P) = P$$

Since P is not normal in G, $\ker \varphi = 1$. That is, φ is indective,

$$G \cong \varphi[G] \leq S_4.$$

Since G Contains 8 elements of order 3, by Counting,

and S_4 has exactly 8 element of order 3, and all these are even,

Thus $G \cong A_4$.

⌈Groups of $p^2 \cdot q$, p and q distinct primes.

Let G be a group of order $p^2 \cdot q$. Let $P \in Syl_p(G)$ and $Q \in Syl_p(G)$.

⌈In case of $p > q$, then $n_p = 1$ or $1+kp$ $(k \geq 1)$, but $1+kp \nmid q$, so $n_p = 1$.

Thus $P \trianglelefteq G$.

「If $|G|=60$ and G has more than one Sylow 5-subgroup, then G is simple.

Proof. $60 = 5!$.. Since $n_5 > 1$. Then only possibility is $n_5 = 6$.

Assume there is non-trivial proper normal $H \subsetneq G$ exists.

If $5 \mid |H|$, then it has at least one sylow-5 subgroup, and since $H \trianglelefteq G$, all sylow 5-subgroups are contained in H, so,

$$|H| \geq 1 + 4 \cdot 6 = 25,$$

therefore $|H| = 30$. but, since order of 30 has a unique normal sylow 5-subgroup, so contradiction. Thus, $5 \nmid |H|$.

If $|H| = 6$ or 12,